Chapter 14, Problem 19P

Problem

Sketch the asymptotic Bode plots of the magnitude and phase for

$$H(s) = \frac{80s}{(s+10)(s+20)(s+40)}, \qquad s = j\omega$$

Step-by-step solution

Step 1 of 9

Consider the transfer function as,

$$H(s) = \frac{80s}{(s+10)(s+20)(s+40)}$$

Replace $j\omega$ for s in equation.

$$\mathbf{H}(\omega) = \frac{80\,j\omega}{(j\omega+10)(j\omega+20)(j\omega+40)}$$

Write the transfer function in the standard form.

$$\begin{aligned}\mathbf{H}(\omega) &= \frac{80\,j\omega}{(10)(20)(40)\left(1+\frac{j\omega}{10}\right)\left(1+\frac{j\omega}{20}\right)\left(1+\frac{j\omega}{40}\right)}\\ &= \frac{(80)(j\omega)}{(8000)\left(1+\frac{j\omega}{10}\right)\left(1+\frac{j\omega}{20}\right)\left(1+\frac{j\omega}{40}\right)}\\ &= \frac{0.01\,j\omega}{\left(1+\frac{j\omega}{10}\right)\left(1+\frac{j\omega}{20}\right)\left(1+\frac{j\omega}{40}\right)}\end{aligned}$$

Step 2 of 9

Calculate the magnitude of the transfer function.

$$H_{\mathrm{dB}} = 20\log 0.01 + 20\log\omega - 20\log\left(1+\frac{\omega^2}{100}\right) - 20\log\left(1+\frac{\omega^2}{400}\right) - 20\log\left(1+\frac{\omega^2}{16000}\right)$$

At $\omega = 0.1\ \mathrm{rad/s}$,

$$\begin{aligned}H_{\mathrm{dB}} &= 20\log(0.01) + 20\log(0.1) - 20\log\left(1+\frac{0.01}{100}\right) - 20\log\left(1+\frac{0.01}{400}\right) - 20\log\left(1+\frac{0.01}{1600}\right)\\ &\simeq 20\log(0.01) + 20\log(0.1)\\ &= -60\,\mathrm{dB}\end{aligned}$$

At $\omega = 10\ \mathrm{rad/s}$,

$$\begin{aligned}|H(j\omega)|_{\mathrm{dB}} &= 20\log(0.01) + 20\log(10) - 20\log\left(1+\frac{100}{100}\right)\\ &= -40 + 20 - 6.02\\ &= -26.02\,\mathrm{dB}\end{aligned}$$

Step 3 of 9

Consider the corner frequencies are $0, 10, 20, 40$.

The transfer function has one simple zero at origin. So the initial slope of the magnitude plot is $+20\ \text{dB/decade}$.

The magnitude plot starts at $-60\,\text{dB}$ with a slope of $+20\ \text{dB/decade}$ and it is extended up to the next corner frequency.

At a corner frequency of 10, one pole is added to the transfer function with a slope of $-20\ \text{dB/decade}$. The net slope of the magnitude plot at $\omega = 10$ is $0\ \text{dB/decade}$.

The magnitude plot continuous with $0\ \text{dB/decade}$ up to the next corner frequency that is $\omega = 20$.

Step 4 of 9

At a corner frequency of 20, one pole is added to the transfer function with a slope of $-20\ \text{dB/decade}$. The net slope of the magnitude plot at $\omega = 20$ is $-20\ \text{dB/decade}$.

The magnitude plot continuous with $-20\ \text{dB/decade}$ up to the next corner frequency that is $\omega = 40$.

At a corner frequency of 40, one pole is added to the transfer function with a slope of $-20\ \text{dB/decade}$. The net slope of the magnitude plot at $\omega = 40$ is $-40\ \text{dB/decade}$.

The magnitude plot continuous with $-40\ \text{dB/decade}$ up to the next corner frequency that is $\omega = \infty$.

Step 5 of 9

Draw the asymptotic magnitude plot as shown in Figure 1.

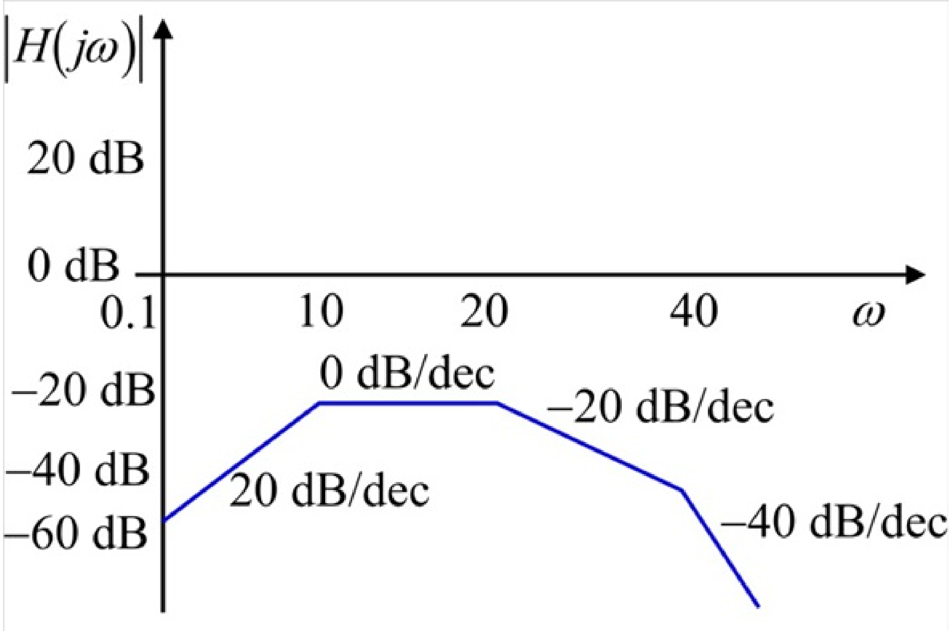

Figure 1

Step 6 of 9

The initial slope of the phase plot is $+90°$.

At $\omega = 10$, one pole is added to the transfer function. So a phase of $-90°$ is added to the phase plot.

The phase at $\omega = 10$ is $0°$ and it is extended up to the next corner frequency $\omega = 20$.

At $\omega = 20$, one more pole is added to the transfer function. So a phase of $-90°$ is added to the phase plot. The phase at $\omega = 20$ is $-90°$ and it is extended up to the next corner frequency $\omega = 40$.

At $\omega = 40$, one more pole is added to the transfer function. So a phase of $-90°$ is added to the phase plot. The phase at $\omega = 40$ is $-180°$ and it is extended up to the next corner frequency $\omega = \infty$.

Step 7 of 9

Draw the asymptotic phase plot as shown in Figure 2.

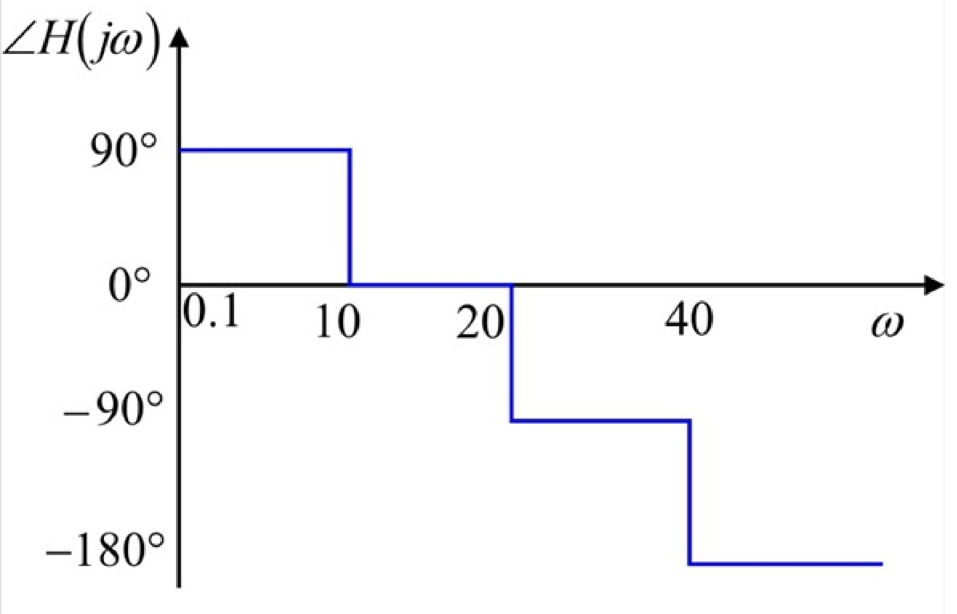

Figure 2

Step 8 of 9

Consider the transfer function.

$$H(s) = \frac{80s}{(s+10)(s+20)(s+40)}$$
$$= \frac{80s}{s^3 + 70s^2 + 1400 + 8000}$$

Consider the MATLAB code to draw the bode plot.

```
num=[80 0];
den=[1 70 1400 8000];
sys=tf(num,den);
bodeplot(sys)
grid
```

Step 9 of 9

The magnitude and phase plots obtained using MATLAB codes are shown in Figure 3.

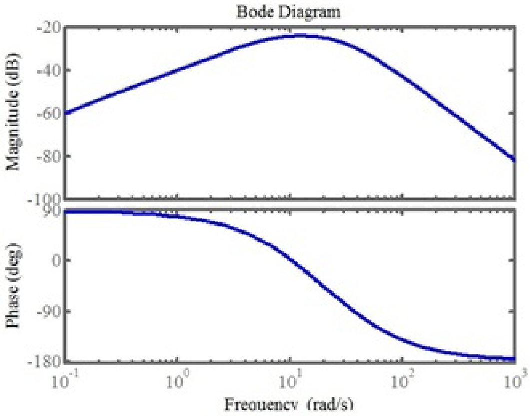

Figure 3